

Angles from intersecting chords, secants and tangents



1

a $x = 103$

c $x = 95, y = 85$

e $x = 74$

g $x = 245, y = 65$

i $x = 180, y = 45, z = 45$

b $x = 123$

d $x = 25$

f $x = 38$

h $x = 60, y = 148$

2

a 110°

b 35°

c 4.88 units

3

a 114°

b 114°

c 33°

d 33°

e 66°

4

a 90°

b 10

c $5\sqrt{3}$

d 36°

e 72°

5

a 13.5°

b 53°

c 100°

d 50°

e 63.5°

Additional questions

6

a $m\widehat{AEB} = 266^\circ$

c $x = 40$

e $m\angle UVT = 114^\circ$

g $m\angle DEB = 81^\circ$

i $x = 43^\circ$

b $m\angle ACE = 40^\circ$

d $x = 64$

f $x = 10$

h $m\widehat{TU} = 130^\circ$

j $m\angle H = 52^\circ$

7

a $x = 7.25$

b $m\angle 1 = 33.75^\circ$



8

a $x = 20.6$

b $m\widehat{ADC} = 172^\circ$

9

a $m\widehat{EC} = 136^\circ$

b $m\angle BEC = 48^\circ$

10 $x = 12$

11

a $m\angle 1 = 28.75^\circ$

b $m\widehat{AB} = 91^\circ$

12

a 120°

b 91°

13 Slices B, C, and D would fit. Slice D would not because the arc has a measure of 146° .

14

a $x = 6$

b $m\angle 1 = 49.5^\circ$

c $m\widehat{EDC} = 125^\circ$

d $m\angle 2 = 61.5^\circ$

15

$\angle KLM$ is formed by a secant and a tangent, so $m\widehat{KL} = 2x^\circ$. $\angle KJL$ intercepts $\widehat{KL}$, so $m\angle KJL = x^\circ$. Since the given triangle is isosceles we know that $m\angle KLJ = x^\circ$, and using triangle sum theorem we get $x = 45$. Since $x = 45$, $m\angle JKL = 90^\circ$, meaning it must be inscribed in a diameter. Therefore $\overline{JL}$ is a diameter of the circle.

16

a Model answer:

b $\angle KLM = \frac{1}{2}\angle N$

